Microeconomics I

Class 6 — Expenditure minimisation and the cost of living

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14 October 2026

Plan for today

Warm-up; two price indices

Expenditure minimisation: h and e

In pairs: kinks and corners

Relabelling; properties of e and h

Break

Shephard's lemma; duality

Compensated price effects; net substitutes

The true cost of living

In pairs: fixed basket vs true index; summary

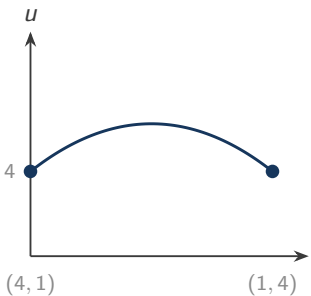
Practicalities

- Problem set 1: on the desk at the start of class, one paper copy per group
- Problem set 2: handed out in class today, one per group;
  due at the start of class, Wed 21 Oct
- Office hours: Wednesdays 14:00–15:00, Room 445

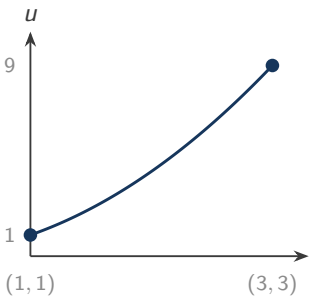
  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Warm-up: concave, or only quasiconcave?

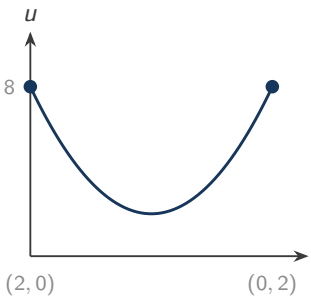
Utility along the segment between two bundles. What does each picture rule out?



x_1x_2 **from** $(4,1)$ **to** $(1,4)$



x_1x_2 **from** $(1,1)$ **to** $(3,3)$



$x_1^3 + x_2^3$ **from** $(2,0)$ **to** $(0,2)$

Class 3, slides 21–22; MWG 3.C, pp. 49–50.

Two US price indices, the same prices: which rises faster?

United States: two official price indices, the same prices

CPI-U: today's prices of an earlier basket (Laspeyres)

Chained CPI-U, since 2002: the basket follows what people buy

Which rises faster, and by how much a year?

0.02 · 0.2 · 2 percentage points (average, 2001–2023)

Official US statistics; the source in class.

Revealed in class

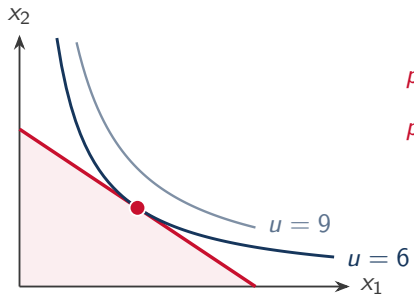
Before class: which should rise faster, and why? Think of one price that doubles.

Expenditure minimisation

MWG 3.E

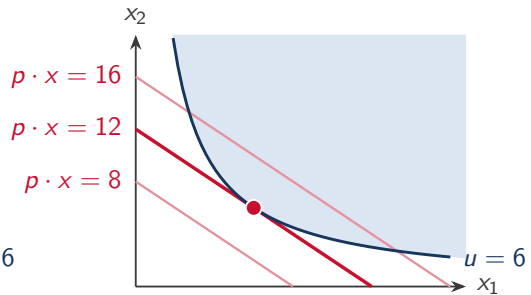
Two problems, one tangency: fix the budget, or fix the curve

EMP: $\min_{x \geq 0} \; p \cdot x \;$ s.t. $\; u(x) \geq u$ value $e(p, u)$, solution $h(p, u)$



UMP: $w = 12$ **fixed**

the highest curve the line reaches: $u = 6$



EMP: $u = 6$ **fixed**

the lowest line that reaches the curve: 12

Same bundle $(3, 2)$: the best bundle 12 euros buy is the cheapest way to reach $u = 6$.

MWG 3.E, p. 58: Figure 3.E.1, Proposition 3.E.1; $u = x_1 x_2$, $p = (2, 3)$.

For x_1x_2: compensated demand and the least cost of u

tangency $x_2/x_1 = p_1/p_2$, on the curve $x_1x_2 = u$:

$$h(p,u) = \left(\sqrt{u\,p_2/p_1},\ \sqrt{u\,p_1/p_2}\right)$$

$$e(p,u) = p \cdot h(p,u) = 2\sqrt{u\,p_1p_2}$$

at $p = (2,3)$, $u = 6$: $h = (3,2)$, $e = 12$

No w anywhere: h and e depend on prices and the target u.

MWG 3.E, pp. 59–61; Example 3.E.1, p. 63, does $x_1^{\alpha}x_2^{1-\alpha}$; ours is x_1x_2.

e and v undo each other: four identities

identity	check for $x_1 x_2$
$e(p, v(p, w)) = w$	$2\sqrt{\dfrac{w^2}{4p_1p_2}\, p_1p_2} = w$
$v(p, e(p, u)) = u$	$\dfrac{e(p, u)^2}{4p_1p_2} = u$
$h(p, u) = x(p, e(p, u))$	$\dfrac{e(p, u)}{2p_1} = \sqrt{u\, p_2/p_1}$
$x(p, w) = h(p, v(p, w))$	$\sqrt{v(p, w)\, p_2/p_1} = \dfrac{w}{2p_1}$

Know v? Invert it in its last argument: that is e.

MWG Prop. 3.E.1, p. 58; (3.E.1), p. 60; (3.E.4), p. 62. $v = w^2/(4p_1p_2)$: class 5, slide 8.

In pairs: the cheapest way to the curve, with kinks and corners

$p = (2, 3)$: the cheapest bundle that reaches u; then $e(p, u)$ for any p.

$u(x)$	reach	cheapest bundle	$e(p, u)$
$\min\{x_1, x_2\}$	$u = 2.4$		
$x_1 + x_2$	$u = 6$		
$x_1^3 + x_2^3$	$u = 216$		

Four minutes.

Answers revealed in class. Try it in pairs before class.

Relabel the curves: the cheapest bundle and its cost stay put

u	$v(p,w)$	$e(p,u)$	the curve through $(3,2)$
x_1x_2	$\dfrac{w^2}{4p_1p_2}$	$2\sqrt{u\,p_1p_2}$	$u=6$: $\;e=12$
$\frac{1}{2}\ln x_1+\frac{1}{2}\ln x_2$	$\ln w-\frac{1}{2}\ln(p_1p_2)-\ln 2$	$2\exp(u)\sqrt{p_1p_2}$	$u=\frac{1}{2}\ln 6$: $\;e=12$

Replace u by $f(u)$: v becomes $f(v)$; e and h keep euros and bundles, only the label changes.

MWG p. 57 ($\tilde{v}=f(v)$) and (3.E.1), p. 60; class 5, slide 12.

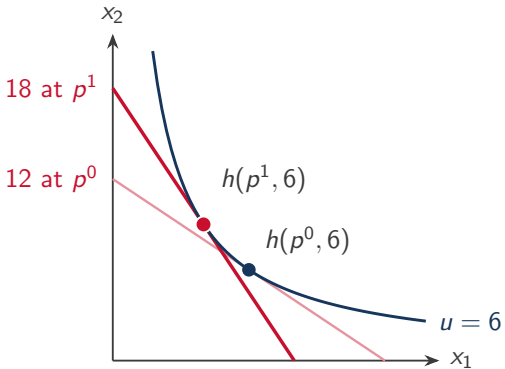
$e(p, u)$: homogeneous of degree 1 in p, increasing in u, concave in p

property	why	for $e = 2\sqrt{u\, p_1 p_2}$
(i) homogeneous of degree 1 in p	same curve, every cost $\times\, \alpha$	$2\sqrt{u\, \alpha p_1\, \alpha p_2} = \alpha\, e$
(ii) strictly increasing in u	continuity of u	$\dfrac{\partial e}{\partial u} = \sqrt{\dfrac{p_1 p_2}{u}} > 0$
nondecreasing in each p_ℓ	no bundle gets cheaper	$\dfrac{\partial e}{\partial p_1} = \sqrt{\dfrac{u\, p_2}{p_1}} > 0$
(iii) concave in p	after the break	
(iv) continuous		

Tuesday's v: degree 0 and quasiconvex. Today's e: degree 1 and concave.

MWG Proposition 3.E.2, p. 59: u continuous, $\succsim$ locally non-satiated, $X = \mathbb{R}^2_+$.

$h(p, u)$ stays on the curve and never rises with its own price



(i) $h(\alpha p, u) = h(p, u)$

only relative prices matter

(ii) $u\Big(h(p, u)\Big) = u$

no utility to spare

(iii) $\succsim$ strictly convex: one bundle

convex: a convex set of bundles

$(p^1 - p^0) \cdot \left(h^1 - h^0\right) \leq 0$

here $(2.5,\ 0) \cdot (-1,\ 1) = -2.5$

Good 1 dearer, utility held at $u = 6$: she buys less of it here, and never more in general.

MWG Prop. 3.E.3, p. 61; Prop. 3.E.4 and (3.E.5), p. 62; $u = x_1 x_2$, $p^0 = (2, 3)$, $p^1 = (4.5, 3)$.

(optional) Both prices move: which bundles can be the new h?

Utility target u fixed. At $p^0 = (2, 3)$ her cheapest bundle is $h^0 = (3, 2)$.
Prices move to $p^1 = (3, 1)$. Could $h(p^1, u)$ be this bundle?
Check slide 14's law and the definition of h, at both price vectors.

(1) $(2,\ 4)$

(2) $(4,\ 1)$

(3) $(2,\ 2.5)$

At home if not done in class; the answers are in the slides released after class.

MWG Proposition 3.E.4 and (3.E.5), p. 62: u continuous, $\succsim$ locally non-satiated, $h(p, u)$ single-valued.

Break

Ten minutes

Shephard's lemma and duality

MWG 3.F, 3.G

Shephard's lemma: compensated demand is the price slope of e

$$h_\ell(p,u) = \frac{\partial e(p,u)}{\partial p_\ell}$$

u continuous; $\succsim$ locally non-satiated and strictly convex

$e = 2\sqrt{u\, p_1 p_2}$: $\quad \dfrac{\partial e}{\partial p_1} = \sqrt{u\, p_2/p_1} = h_1$

slide 9

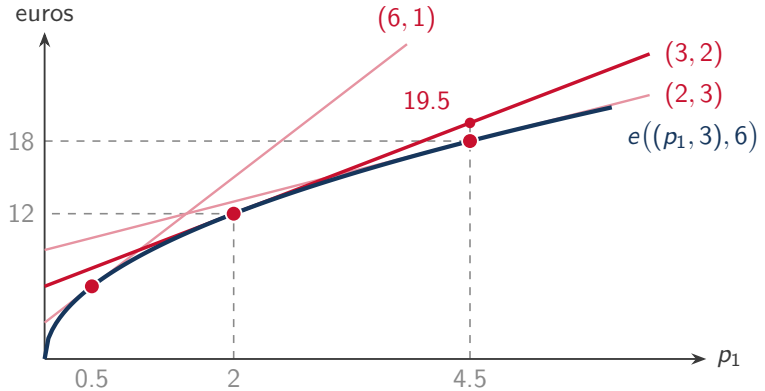
Roy (class 5): $\quad x_\ell = -\dfrac{\partial v/\partial p_\ell}{\partial v/\partial w}$ $\qquad\qquad$ Shephard: $\quad h_\ell = \dfrac{\partial e}{\partial p_\ell}$

Roy divides to turn utils into euros; e is in euros already.

MWG Proposition 3.G.1 (Shephard's lemma), p. 68; Roy's identity: Prop. 3.G.4, p. 73.

e is the lowest of straight lines: concave in p, slope $= h$

$u = x_1 x_2$, $u = 6$, $p_2 = 3$: each straight line is the cost $p_1 x_1 + 3x_2$ of one bundle on the curve



At $p_1 = 2$ the line of $(3, 2)$ touches e with slope $3 = h_1$: Shephard's lemma, drawn.

MWG Figure 3.E.2, p. 60; Figure 3.F.3 and the duality theorem (Prop. 3.F.1), pp. 65–66.

Compensated price effects: own effects ≤ 0, cross effects equal

$$D_p h(p,u) = \begin{pmatrix} \partial h_1/\partial p_1 & \partial h_1/\partial p_2 \\ \partial h_2/\partial p_1 & \partial h_2/\partial p_2 \end{pmatrix} = \begin{pmatrix} -\frac{3}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{3} \end{pmatrix}$$

$x_1 x_2$ at $p = (2,3)$, $u = 6$

own effects ≤ 0

negative semidefinite

$\partial h_1/\partial p_2 = \partial h_2/\partial p_1$

symmetric

$D_p h\ p = 0$: $\quad -0.75 \cdot 2 + 0.5 \cdot 3 = 0$

degree 0 in p

Two goods: $\partial h_1/\partial p_2 = -(p_1/p_2)\,\partial h_1/\partial p_1 \geq 0$.

MWG Proposition 3.G.2, p. 69: $D_p h(p,u) = D_p^2 e(p,u)$.

Net substitutes: with two goods, never net complements

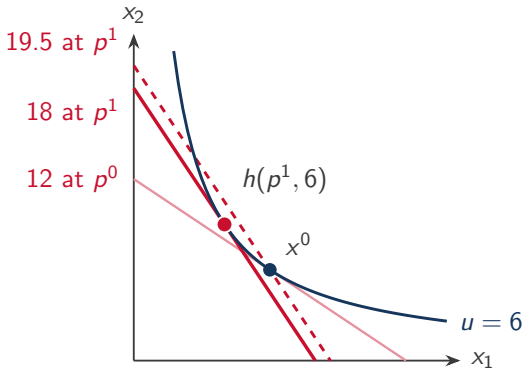
net substitutes: $\partial h_1/\partial p_2 > 0$; net complements: < 0

σ	u	gross: x_1, p_2: $3 \to 4$	net: h_1, p_2: $3 \to 4$
	$p_1 = 2$; gross at $w = 12$ (class 5); net at the utility $w = 12$ bought		
0	$\min\{x_1, x_2\}$	$2.4 \to 2$: complements	$2.4 \to 2.4$: neither
1	$x_1 x_2$	$3 \to 3$: neither	$3 \to 3.46$: substitutes
2	CES, $\rho = \frac{1}{2}$	$3.6 \to 4$: substitutes	$3.6 \to 4.44$: substitutes

Gross and net can disagree; the gap is a wealth effect: next Tuesday.

MWG 3.G, p. 70 (weak inequalities; strict here, so zero is "neither"). Class 5, slide 23.

True cost of living: $\times 1.5$; the fixed basket says $\times 1.625$



true index: $\dfrac{e(p^1, u^0)}{e(p^0, u^0)} = \dfrac{18}{12} = 1.5$

fixed basket (Laspeyres):

$$\frac{p^1 \cdot x^0}{p^0 \cdot x^0} = \frac{19.5}{12} = 1.625$$

x^0 still reaches u^0, so $e(p^1, u^0) \leq p^1 \cdot x^0$: never below the truth

To stay on her curve she needs 6 more euros, not 7.5.

Konüs (1924; Econometrica 1939): the true index. Own numbers: $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (4.5, 3)$.

In pairs: good 1 now costs 8: how much richer must each be?

$p^0 = (2, 3)$, $w = 12 \;\rightarrow\; p^1 = (8, 3)$. Fixed-basket index and true index.

$u(x)$	x^0	fixed basket $\frac{p^1 \cdot x^0}{12}$	true $\frac{e(p^1, u^0)}{12}$
$\min\{x_1, x_2\}$	(2.4, 2.4)		
$x_1 x_2$	(3, 2)		
$x_1 + x_2$	(6, 0)		

Seven minutes.

Answers revealed in class. Try it in pairs before class.

(optional) Price her new basket: how does that index compare?

Slide 22's consumer: $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (4.5, 3)$.
At p^1 she buys x^1, with utility u^1.

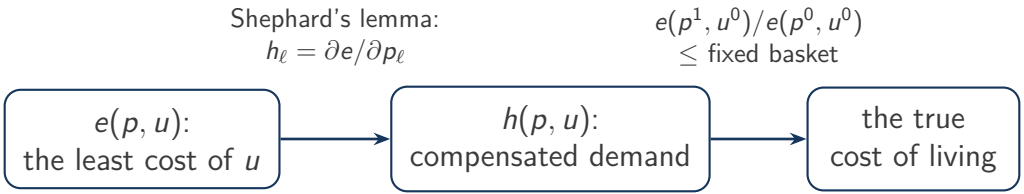
(1) x^1, and the Paasche index $P = \dfrac{p^1 \cdot x^1}{p^0 \cdot x^1}$.

(2) Show, under the assumptions of
Prop. 3.E.1: $P \leq \dfrac{e(p^1, u^1)}{e(p^0, u^1)}$.

(3) Order P and slide 22's two indices.

At home if not done in class; the answers are in the slides released after class.

Today in one line



Tuesday: the Slutsky equation and the Slutsky matrix; summary of the relationships; compensating and equivalent variation.

Problem set 2: handed out in class today, one per group; due at the start of class, Wednesday 21 October, on paper.

Read MWG 3.G and 3.I (or Muñoz-Garcia ch. 2) for Tuesday 20 October.